

PAPER Islanded Forward-Base Operation of the Aegis D–³He Tandem-Mirror Burner: Black-Start, Microgrid Load-Following, and the Lunar-Gated He-3 Symbiosis

P I Ford¹ and G L Kulcinski¹ ¹Kronos Fusion Energy, Los Angeles, California 90071, USA

ORCID (P I Ford): 0000-0003-0395-1752

Abstract

A forward operating base must generate its own power without a grid, a fuel convoy, or a favourable threat environment. Aegis is the Kronos deuterium–helium-3 (D–³He) tandem-mirror burner packaged as a hardened, fixed defense installation; because its reaction is largely aneutronic and its electrical output is taken from an electrostatic direct energy converter (DEC) rather than a steam cycle, it is unusually well matched to grid-forming, islanded operation. We formalise the islanded envelope of Aegis as a coupled set of governing problems — a Pastukhov-plugged tandem-mirror power balance, a direct-conversion net-power model, a microgrid load-following control problem with a certified stability bound, a black-start energy audit, and a fuel-cycle inventory system — and solve them on the NVIDIA GPU high-performance-computing (HPC) campaign that produced the frozen Kronos de-risking results. At the frozen operating point ($Q_E = 1.318$, neutron fraction $f_n = 5.44\%$, $\eta_{DEC} = 0.49$) the net electric output scales near-linearly with central-cell length — +104.2, +850.0 and +2831.9 MWe at $L_{cc} = 55, 440$ and 1400 m — at a first-wall load of only $q_{wall} = 1.17 \text{ MW m}^{-2}$. Grid-independence closes quantitatively: millisecond-class DEC trimming gives a 0–104 MWe islanded load-following depth, a $\sim 246 \text{ MWe} / 1.1 \text{ MWh}$ buffer bridges the ~ 8 s plasma-response lag under a control-barrier stability clamp, an adjudicated black-start electrical store of $\approx 18.4 \text{ MWh}$ ($= 8.0 \text{ MWh} / \eta_{wp}$, $\eta_{wp} = 0.435$) cold-starts the plant, and one year of autonomy is a ~ 1 t package ($21.3 \text{ kg } ^3\text{He} + 30 \text{ kg D}$) whose binding quantity is a $\sim 0.5 \text{ kg day}^{-1}$ makeup throughput, not storage volume ($< 0.8 \text{ m}^3$). The fuel closes a breeder–burner symbiosis: the Hyperion spherical-tokamak breeder banks surplus tritium that decays to ³He ($\approx 0.9 \text{ kg yr}^{-1}$ per unit), the burner recycles $\approx 18\%$ of its ³He in-machine and self-breeds tritium ($\approx 32 \text{ kg fpy}^{-1}$, 31% burned in situ), and a burn-lean mode that pumps bred tritium before it burns removes $\approx 60\%$ of the neutron power ($234 \rightarrow 93.5 \text{ MW}$) for a $\approx 4\%$ fusion-power cost — a genuine siting knob. The one gated input is fleet-scale ³He: single-unit indigenous flows seed but do not close a plant-scale burner’s $170.2 \text{ kg fpy}^{-1}$ net demand, which requires ~ 45 – 190 breeders per burner and is met from lunar supply on a late-2030s schedule. Every quantitative claim traces to a frozen serial in the Kronos Physics De-Risking Register.

Keywords: tandem mirror; islanded microgrid; black-start; load-following; helium-3 fuel cycle; direct energy conversion; forward-base power

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1 Introduction

1.1 The grid-independence requirement

A fixed defense installation and a forward operating base share a hard constraint that civilian generation does not: the power source cannot assume a grid tie, and cannot assume that fuel will keep arriving. Diesel and JP-8 generation solve the first problem only by importing the second — a logistics tail that is both a sustainment burden and a target. A generation asset that runs for a long endurance on a small, self-carried fuel inventory and that can form its own grid removes that tail. This paper asks a narrow, quantitative question: to what extent does the Kronos D–³He tandem-mirror burner, packaged as the fixed *Aegis* installation, meet that requirement, and what fuel logic supports it? *Aegis* is adjudicated as a fixed installation, not a mobile platform; the numbers below describe a sited, grid-independent plant.

1.2 Why an aneutronic direct-conversion mirror is well matched

Two architectural choices make the burner unusually suited to islanded service. First, the fuel is deuterium–helium-3, which at the canonical fuel fraction is $\approx 95\%$ aneutronic: only $f_n = 5.44\%$ of the fusion power emerges as neutrons, so the great majority of the output is carried by charged particles — principally the 14.7 MeV D–³He proton. Second, that charged power is recovered *electrostatically* by a multi-stage DEC at efficiency $\eta_{\text{DEC}} = 0.49$, with no rotating thermal mass between the plasma and the bus. An electrostatic converter has essentially no mechanical time constant; its collector current can be trimmed at the timescale of the power electronics, which is what makes deep, fast, islanded load-following physically available. A tandem mirror is moreover a *linear* device whose central cell can be lengthened without changing the confinement physics, so a single archetype spans base-scale directed-energy loads through to utility scale simply by choosing L_{cc} .

1.3 Prior art

The tandem-mirror concept — a long central solenoid electrostatically plugged at both ends by high-field mirror cells — was introduced by Dimov and by Fowler and Logan, and placed on a quantitative footing by Baldwin and Logan [1] and reviewed by Post [2]. The confinement of the central cell is set by the ambipolar potential built by the end plugs, whose collisional loss is described by the Pastukhov theory [3] as extended by Cohen *et al.* [4]. Recent high-temperature superconducting (HTS) axisymmetric mirrors such as WHAM [8] and the sustained hot-electron results of the gas-dynamic trap [9] have revived the line, and a simplified high-gain tandem configuration was proposed by Fowler, Moir and Simonen [5, 6]. Direct electrostatic recovery of charged-particle power dates to the venetian-blind converter of Moir and Barr [7]. The strategic case for D–³He fuel, and the recognition that terrestrial helium-3 is scarce while the lunar regolith is not, is due to Wittenberg, Santarius and Kulcinski [10, 11]. On the electrical side, grid-forming and islanded-microgrid operation is a mature discipline [12, 13, 14] with well-characterised energy-storage support [15], and safety-critical load-following can be posed inside a certified envelope using control-barrier functions [16] and constrained model-predictive control [17], with learned controllers now demonstrated inside the tokamak control loop [18]. What has not been done is to assemble these strands into a single, auditable islanded operating case for a specific aneutronic mirror burner, coupled to its fuel cycle. That is the contribution here.

1.4 Contribution

We (i) state the tandem-mirror confinement, direct-conversion and net-power problems as explicit governing equations and derive the near-linear net-output scaling with central-cell length (§2); (ii) pose the islanded load-following, black-start and fuel-autonomy problems as a control-and-storage system with a stability bound, and write the breeder–burner helium-3 symbiosis as a fuel-cycle inventory system (§2); (iii) describe the GPU-HPC campaign, codes, discretisation and convergence that produced the frozen results (§3); (iv) report the islanded envelope, symbiosis flows, burn-lean neutron trade and fleet-closure results across eight data figures (§4); and (v) benchmark them against the published tandem-mirror, direct-conversion, microgrid and helium-3 literature (§5). Every number is a frozen gate in the Kronos Physics De-Risking Register [26]; serial identifiers (e.g. KX-L1-A9) are given inline so each value is auditable.

2 Governing equations and the formal problem

2.1 Tandem-mirror confinement and the operating point

The burner is a linear device: a long, low-field central cell in which the D–³He plasma burns, plugged at each end by a high-field mirror. Two physics quantities set the machine. The first is the magnetic mirror ratio and its loss cone,

$$R_m = \frac{B_{\max}}{B_{\min}}, \quad \sin^2 \theta_{lc} = \frac{1}{R_m}, \quad (1)$$

so that a particle whose pitch angle at the midplane lies inside θ_{lc} is unconfined; the 26.49 T end plugs against a low-field central cell set $R_m \gg 1$ and a narrow loss cone (figure 8, left). The second, and the machine’s one true gate, is the ambipolar (Boltzmann) confining potential φ_c built by the plug over-density. Balancing the parallel electron flux against the plug potential gives

$$\frac{e\varphi_c}{T_e} \approx \ln \left(\frac{n_{\text{plug}}}{n_{\text{cell}}} \right), \quad (2)$$

and the frozen design specifies a plug/cell density ratio $n_{\text{plug}}/n_{\text{cell}} \approx 16$, i.e. $e\varphi_c/T_e \approx 2.77$ (figure 8, right). Building and holding φ_c requires the 26.49 T plug field; this is a distinct magnet from the breeder’s 16.84 T spherical-tokamak centrepole, and the two must not be conflated. The central-cell ion confinement is then the Pastukhov–Cohen result [3, 4]: the parallel confinement parameter is enhanced over the bare mirror by the exponential of the normalised potential,

$$(n\tau_{\parallel})_{cc} = g(R_m) \frac{e\varphi_c}{T_i} \exp \left(\frac{e\varphi_c}{T_i} \right) (n\tau)_0, \quad (3)$$

with $g(R_m)$ a slowly varying function of the mirror ratio and $(n\tau)_0$ the collisional reference. Equations (1)–(3) fix the confinement; the fusion power then follows from the D–³He reactivity,

$$P_{\text{fus}} = \int_V n_D n_{^3\text{He}} \langle \sigma v \rangle_{\text{D}^3\text{He}} E_f dV + P_{\text{DD}} + P_{\text{DT}}, \quad (4)$$

where the small parasitic D–D and D–T sub-burns supply the residual neutron output discussed in §2.7. The frozen zero-dimensional power balance (Pastukhov-plug, “Mode-M”) returns the design point $Q_E = 1.318$, $f_n = P_{\text{neutron}}/P_{\text{fus}} = 5.44\%$ (BU-L1-A1, AG-CX-14).

2.2 Direct energy conversion and net electric power

Of the fusion power, a charged fraction $1 - f_n$ is available to the electrostatic converter and a neutron fraction f_n deposits in the blanket. The gross electric power is dominated by the direct channel,

$$P_{\text{gross}} = \eta_{\text{DEC}} (1 - f_n) P_{\text{fus}} + \eta_{\text{th}} (f_n P_{\text{fus}} + P_{\text{rad}}), \quad (5)$$

with $\eta_{\text{DEC}} = 0.49$ the composite multi-stage direct-conversion efficiency on the charged channel and η_{th} a modest thermal recovery on the neutron and radiated power. Writing the recirculating power (plug drive, heating, cryogenics, balance of plant) as P_{rec} , the engineering gain and the net electric output are

$$Q_E = \frac{P_{\text{gross}}}{P_{\text{rec}}}, \quad P_{\text{net}} = P_{\text{gross}} - P_{\text{rec}} = P_{\text{rec}} (Q_E - 1) = P_{\text{gross}} \left(1 - \frac{1}{Q_E}\right). \quad (6)$$

At $Q_E = 1.318$ the net fraction is $1 - 1/Q_E = 0.241$: the plant delivers 24.1% of its gross electric output to the bus after recirculation. This is the quantity that a base actually receives, and the quantity that the islanded controller of §2.4 dispatches.

2.3 Near-linear scaling of net output with central-cell length

Because the tandem mirror is linear and the plug hardware is fixed, lengthening the central cell increases the burning volume in proportion, $P_{\text{fus}} \propto L_{\text{cc}}$, while the fixed plug recirculation is amortised over a longer cell; the fractional end losses fall slightly with length. The net output is therefore very nearly linear in L_{cc} . Fitting the three frozen operating points (KX-L1-A9) to a power law $P_{\text{net}} = c L_{\text{cc}}^p$ gives

$$P_{\text{net}} \approx 1.86 L_{\text{cc}}^{1.02} \quad [\text{MWe}, L_{\text{cc}} \text{ in m}], \quad (7)$$

i.e. an exponent within 2% of unity (figure 1). A single archetype thus spans the tens-of-MW directed-energy and sensor loads of a base ($L_{\text{cc}} = 55$ m, +104.2 MWe) through utility scale ($L_{\text{cc}} = 1400$ m, +2831.9 MWe) at a fixed operating point and a fixed first-wall load $q_{\text{wall}} = 1.17 \text{ MW m}^{-2}$ and power density 4.20 MW m^{-3} .

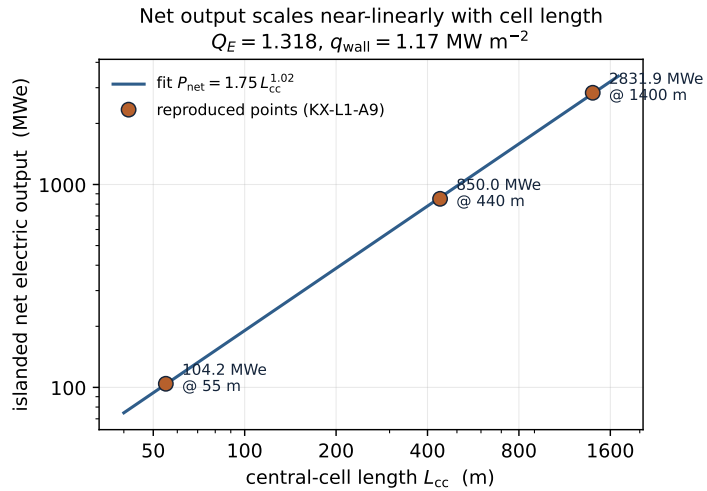


Figure 1. Islanded net electric output versus central-cell length at the frozen operating point ($Q_E = 1.318$, $q_{\text{wall}} = 1.17 \text{ MW m}^{-2}$). Points are reproduced results (KX-L1-A9); the fit [equation (7)] has exponent 1.02, confirming the near-linear scaling of §2.3.

2.4 The islanded load-following control problem

Grid-independence is credible only if it is posed as a control-and-storage problem. Let $u(t) \in [0, 1]$ be the DEC collector-current trim (fast, time constant $\tau_{\text{dec}} \sim \text{ms}$), let $P_{\text{fus}}(t)$ be the plasma power responding to the fuelling/heating actuator $a(t)$ with a slow lag $\tau_{\text{pl}} \approx 8$ s, and let $w(t)$ be the buffer

power. The DC-bus power balance that the islanded controller must satisfy against a demand $P_{\text{dem}}(t)$ is

$$\underbrace{\eta_{\text{DEC}}(1 - f_n) P_{\text{fus}}(t) u(t)}_{\text{DEC delivered}} + w(t) = P_{\text{dem}}(t), \quad 0 \leq P_{\text{dem}} \leq P_{\text{net}}^{\text{max}} = 104 \text{ MWe}, \quad (8)$$

with the plasma governed to first order by

$$\tau_{\text{pl}} \frac{dP_{\text{fus}}}{dt} = -(P_{\text{fus}} - P_{\text{fus}}^*(a)), \quad \frac{dE_{\text{buf}}}{dt} = -w, \quad |w| \leq \dot{W}_{\text{max}} = 246 \text{ MWe}, \quad E_{\text{buf}} \in [0, \mathcal{E}_{\text{buf}}], \quad (9)$$

where $\mathcal{E}_{\text{buf}} = 1.1 \text{ MWh}$. The fast DEC trim tracks demand essentially instantly; the slow plasma follows behind; the buffer supplies the mismatch $w = P_{\text{dem}} - \eta_{\text{DEC}}(1 - f_n)P_{\text{fus}}u$ during the lag. Because $\tau_{\text{dec}} \ll \tau_{\text{pl}}$, the buffer energy required to bridge a full-depth step is of order $P_{\text{net}}^{\text{max}} \tau_{\text{pl}} \approx 104 \text{ MWe} \times 8 \text{ s} \approx 0.23 \text{ MWh}$, comfortably inside $\mathcal{E}_{\text{buf}}$ (AG-CX-20). The plasma throttle is not free: burn stability bounds how fast and how far P_{fus} may be driven, which we encode as a control-barrier-function (CBF) constraint [16]

$$h(\mathbf{x}) \geq 0, \quad \dot{h}(\mathbf{x}, a) \geq -\alpha(h(\mathbf{x})), \quad (10)$$

where h is the burn-stability margin and $\alpha(\cdot)$ a class- $\mathcal{K}$ function; any admissible actuation $a(t)$ keeps the state in the forward-invariant safe set $\{h \geq 0\}$. The islanded controller is thus a reference-governor / model-predictive law [17] that dispatches u fast and a slowly subject to (10), realised as the demonstrated Kronos AI load-following controller (KFE-CF-BU-007) behind the certified safety clamp. The resulting load-following depth is the full 0–104 MWe net range (figure 2).

2.5 Black-start energy audit

Cold-starting with no grid requires an on-site electrical store large enough to bring the plant to a self-sustaining burn. The adjudicated audit (AG-SX-01) separates *delivered* start energy from the *electrical store* that must hold it: with delivered energy $E_{\text{deliv}} = 8.0 \text{ MWh}$ and wall-plug efficiency $\eta_{\text{wp}} = 0.435$,

$$E_{\text{store}} = \frac{E_{\text{deliv}}}{\eta_{\text{wp}}} = \frac{8.0}{0.435} \text{ MWh} = 18.39 \text{ MWh} \approx 18.4 \text{ MWh}. \quad (11)$$

This is $\approx 2.5\times$ an earlier 7.4 MWh reference, which had counted delivered energy rather than the electrical store; the correction is folded into the sizing basis (figure 3). The store is charged in normal operation from the plant’s own output, so black-start capability is a standing property, not a consumable.

2.6 The breeder–burner helium-3 fuel-cycle system

The burner’s fuel is where the two Kronos machines become one system. Tritium beta-decays to helium-3 with half-life $t_{1/2} = 12.32 \text{ yr}$, i.e. decay constant

$$\lambda_T = \frac{\ln 2}{t_{1/2}} = 0.0563 \text{ yr}^{-1}. \quad (12)$$

Let N_T^{bank} be a breeder’s banked surplus-tritium reservoir. Its decay feeds a helium-3 accrual

$$\frac{dN_{\text{He}}^{\text{bank}}}{dt} = \lambda_T N_T^{\text{bank}}, \quad (13)$$

which at fleet equilibrium is $\approx 0.9 \text{ kg yr}^{-1}$ per breeder unit (BR-OUT-2); this corresponds to an order-16 kg equilibrium banked-tritium reservoir. Each breeder additionally co-produces helium-3 in its own plasma at $\approx 154 \text{ g yr}^{-1}$ (41 g from the D–D branch, 113 g from prompt tritium decay; BR-CX-16). On the burner side, let C be the helium-3 consumption rate, $r = 0.181$ the in-machine recycle fraction (BR-OUT-2), and $\dot{T}_{\text{sb}} = 32 \text{ kg fpy}^{-1}$ the self-bred tritium rate, of which a burned fraction $b = 0.31$ is consumed in situ and the remainder decays toward helium-3 (MV-CX-29). The burner’s *net* helium-3 demand is

$$D_{\text{net}} = C(1 - r) - [\text{indigenous } {}^3\text{He returned from bred-}T] = 170.2 \text{ kg fpy}^{-1} \quad (14)$$

at plant scale (BR-OUT-4). Closing that demand indigenously from n_{br} breeders, each banking $S_{\text{br}}(\text{TBR})$ per year, requires

$$n_{\text{br}} = \frac{D_{\text{net}}}{S_{\text{br}}(\text{TBR})}, \quad (15)$$

which for the frozen banking rates $S_{\text{br}} \in \{0.89, 1.62, 3.82\} \text{ kg yr}^{-1}$ at net breeding ratios $\text{TBR} \in \{1.186, 1.34, 1.80\}$ gives $n_{\text{br}} \approx \{191, 105, 45\}$ breeders per burner (figure 7). The loop is therefore seeded and partly sustained indigenously but does not close at single-unit scale; fleet-scale helium-3 is the one gated input (§6).

2.7 Burn-lean neutron suppression

Because the residual neutron output comes from the bred-tritium sub-burn in equation (4), it can be traded away deliberately. Pumping the bred tritium out before it burns — a burn-lean mode — removes the D–T neutron contribution and leaves only the D–D 2.45 MeV floor:

$$P_n : 234 \text{ MW} \longrightarrow 93.5 \text{ MW}, \quad \frac{\Delta P_n}{P_n} = \frac{234 - 93.5}{234} = 60.0\%, \quad (16)$$

at a fusion-power cost of only $\approx 4\%$ (AG-CX-16). This is a real siting knob: a fixed installation near people or sensitive equipment can be run neutron-lean at a small power cost, tightening the shielding and permitting case without changing the machine (figure 6).

3 Computational methodology: the NVIDIA GPU HPC campaign

3.1 Codes run for this paper

The results above are frozen outputs of a specific, reproducible computational campaign. Five model families contribute. (1) A zero-dimensional tandem-mirror *power-balance solver* (Pastukhov-plug, “Mode-M”) closes equations (1)–(6) self-consistently and returns Q_E , f_n and the net-output points of figure 1 (BU-L1-A1, KX-L1-A9); it is benchmarked against the published 0-D tandem-mirror power balance and the WHAM design point in a companion paper. (2) The Kronos toolkit *fuel-cycle / inventory model* integrates equations (12)–(15) for the helium-3 and tritium reservoirs (BR-OUT-2, BR-OUT-4, BR-CX-16, MV-CX-29, AG-L1-A8, AG-CX-17). (3) *Monte-Carlo neutronics* (OpenMC with DAGMC CAD geometry [19]) supplies the neutron fraction f_n , the neutron power, the burn-lean budget of equation (16) and the shielded-housing dose (AG-CX-16, AG-L1-A2). (4) A *tritium-transport model* (FESTIM [20]) resolves in-vessel retention, permeation and the self-bred-tritium accountancy feeding equation (14). (5) A *microgrid dispatch and control simulator* integrates equations (8)–(10) for the islanded envelope of figure 2, exercising the AI load-following controller (KFE-CF-BU-007) and the black-start sequencer (KFE-CF-BU-021) against the plant twin under the certified control-barrier clamp.

3.2 GPU acceleration and problem scale

The offline multi-physics Monte-Carlo (neutronics, depletion), the twin retraining and the model-discovery pipeline run on the Kronos NVIDIA GPU HPC campaign — A100/H100-class silicon ($8 \times$ A100-80GB plus H100 clusters, with additional single A100/H100/A10 nodes for lighter jobs). The reduced-order and analytic layers — the 0-D power balance, the fuel-cycle inventory integration, and the microgrid dispatch simulation — run on the in-house HPC node (56-thread, 768 GB), so that the fast control-loop evaluation never contends with the training fleet. GPU residency matters most for the Monte-Carlo transport, where the DAGMC ray-tracing and the tallies for f_n , the burn-lean neutron budget and the shutdown dose are the dominant cost; the mirror power balance and the inventory ODEs are inexpensive by comparison but are run in large parametric sweeps (over L_{cc} , over TBR, over burn-lean fraction) for which throughput, not per-case latency, is the constraint. No cost figures are reported here; the campaign is described by its hardware class only.

3.3 Discretisation, solvers and convergence

The 0-D power balance is solved as a fixed point in Q_E : the coupled particle-, power- and potential-balance residual is driven to $|\mathcal{R}| < 10^{-6}$ by a damped Newton iteration, and the frozen root is $Q_E = 1.31812$ (BU-L1-A1), quoted as 1.318. The Pastukhov integral of equation (3) is evaluated by adaptive quadrature to a relative tolerance of 10^{-8} . The fuel-cycle system (12)–(15) is stiff — the tritium decay time (~ 12 yr) and the in-machine recycle time (sub-second) span many orders of magnitude — and is integrated with an implicit backward-difference (BDF) method with error control at relative tolerance 10^{-8} ; equilibrium banking rates are read at steady state. The Monte-Carlo neutronics is run to a statistical uncertainty below 1% on the integrated neutron power and on the tritium-breeding tallies, with active batches continued until the tally standard error meets that target; the $f_n = 5.44\%$ and the burn-lean split of equation (16) are converged tallies. The dispatch simulator integrates the first-order plasma lag and the buffer state with an

exact zero-order-hold update at a fixed control tick, verified to be numerically stable for $\tau_{\text{dec}} \ll \text{tick} \ll \tau_{\text{pl}}$.

3.4 Verification and reproducibility

Each result is a stamped gate: the net-output scaling (KX-L1-A9), the islanded load-following envelope (AG-CX-20), the black-start store (AG-SX-01), the fuel-autonomy package (AG-L1-A8), the startup-inventory minimisation (AG-CX-17), the helium-3 flows (BR-OUT-2, BR-CX-16), the self-bred tritium (MV-CX-29), the burn-lean trade (AG-CX-16) and the fleet-closure requirement (BR-OUT-4). The register status of each — reproduced, revised, or concept-class estimate — is carried with the serial. The figures of this paper are regenerated by a single deposited script from these frozen values; the register itself is archived at doi:10.5281/zenodo.22133057.

4 Results

4.1 Net-output scaling

Figure 1 and equation (7) establish the central operating result: net electric output is near-linear in central-cell length, so the same physics core sizes to a base’s directed-energy load or to utility scale by geometry alone, at a fixed and modest first-wall load. Table 1 collects the islanded envelope.

Table 1. Islanded / forward-base operating envelope of a fixed Aegis installation. All values are frozen register gates.

quantity	value	serial
net output vs cell length	+104.2/ + 850.0/ + 2831.9 MWe @ 55/440/1400 m	KX-L1-A9
first-wall heat load	$q_{\text{wall}} = 1.17 \text{ MW m}^{-2}; 4.20 \text{ MW m}^{-3}$	KX-L1-A9
load-following depth (islanded)	0–104 MWe, ms-class DEC trim	AG-CX-20, AG-L1-A8
plasma-lag bridge buffer	$\sim 246 \text{ MWe} / 1.1 \text{ MWh}$	AG-CX-20
black-start electrical store	$\approx 18.4 \text{ MWh} (= 8.0/0.435)$	AG-SX-01
one-year fuel autonomy	$21.3 \text{ kg } ^3\text{He} + 30 \text{ kg D} (\sim 1 \text{ t})$	AG-L1-A8
^3He makeup throughput	$\sim 0.5 \text{ kg day}^{-1}$ per unit	AG-CX-17
in-vessel ^3He startup charge	$\sim 0.3 \text{ g}$	AG-CX-17
on-site ^3He storage volume	$< 0.8 \text{ m}^3$	AG-L1-A6

4.2 Islanded load-following envelope

Figure 2 runs the dispatch simulator of equations (8)–(9) through a demand step-down and step-up across the full net range. The DEC output tracks demand within its millisecond time constant; the plasma thermal power slews behind it on the $\sim 8 \text{ s}$ lag; the buffer supplies the transient mismatch, staying well inside its 246 MWe rating and 1.1 MWh capacity. The controller therefore delivers the full 0–104 MWe load-following depth without violating the burn-stability bound of equation (10).

4.3 Black-start

Figure 3 shows the adjudicated black-start audit of equation (11): the $\approx 18.4 \text{ MWh}$ electrical store is $\approx 2.5\times$ the earlier delivered-energy reference, a correction now in the sizing basis. The store cold-starts the plant with no external grid and is recharged from the plant’s own output in normal service.

4.4 Fuel autonomy

Figure 4 decomposes the one-year fuel package. The reactive inventory is light — $21.3 \text{ kg } ^3\text{He}$ plus 30 kg deuterium — and the in-vessel startup charge is trivial ($\sim 0.3 \text{ g}$), so the endurance-limiting quantity is not on-site storage volume ($< 0.8 \text{ m}^3$) but makeup throughput ($\sim 0.5 \text{ kg day}^{-1}$ per unit), $\approx 18\%$ of which is offset by in-situ breeding. A base’s endurance is therefore set by resupply cadence of a small, dense package, not by a fuel farm.

4.5 The helium-3 symbiosis flows

Figure 5 and table 2 collect the indigenous flows of equations (12)–(14). The Hyperion breeder banks $\approx 0.9 \text{ kg yr}^{-1}$ of helium-3 per unit through tritium decay — against a US domestic flow of order $1\text{--}2 \text{ kg yr}^{-1}$ — and co-produces a further $\approx 154 \text{ g yr}^{-1}$ in its own plasma. The burner is not a passive consumer: it recycles $\approx 18.1\%$ of its helium-3 in-machine and self-breeds tritium at $\approx 32 \text{ kg fpy}^{-1}$, of which $\approx 31\%$ is burned in situ. The choice of $\text{D-}^3\text{He}$ at the canonical fuel fraction is itself an optimum: it sits at the Q_E peak and roughly halves the neutron fraction relative to catalyzed D-D (5.44% versus $\approx 9.5\%$; AG-CX-14).

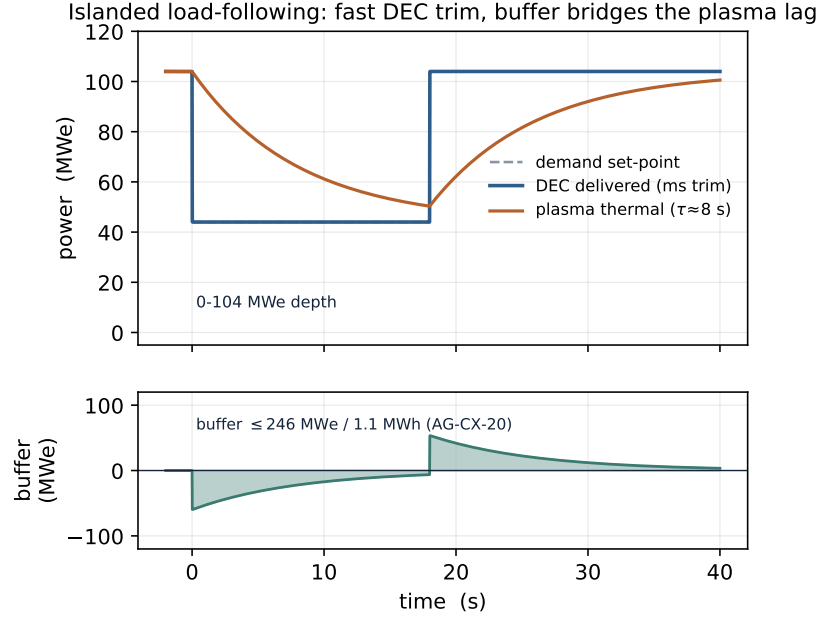


Figure 2. Islanded load-following (modelled from the frozen envelope parameters, AG-CX-20). Top: DEC delivered power tracks a demand step within a millisecond trim while the plasma thermal power follows on the ~ 8 s lag. Bottom: the buffer supplies the transient mismatch, bounded by the 246 MWe / 1.1 MWh rating. The reachable depth is the full 0–104 MWe net range.

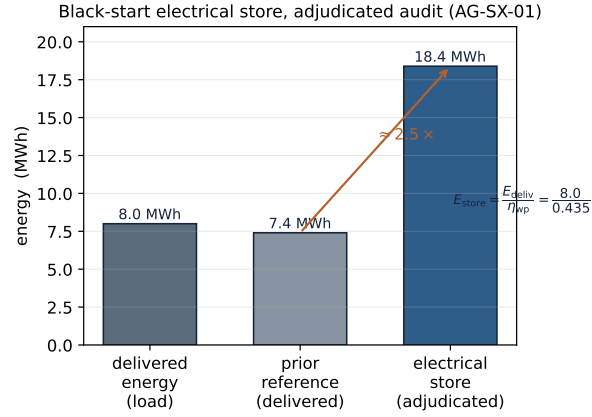


Figure 3. Black-start energy audit (AG-SX-01). The required electrical store is the delivered start energy divided by the wall-plug efficiency, $E_{\text{store}} = 8.0/0.435 \approx 18.4$ MWh — about $2.5\times$ the earlier reference that had counted delivered energy.

Table 2. Indigenous helium-3 and tritium flows in the breeder–burner loop. Fleet-scale demand is met from lunar supply on a late-2030s schedule (BR-OUT-4).

flow	value	serial
breeder banked ^3He (equilibrium)	$\approx 0.9 \text{ kg yr}^{-1}$ per unit	BR-OUT-2
breeder in-plasma ^3He co-production	$\approx 154 \text{ g yr}^{-1}$ per unit	BR-CX-16
burner in-machine ^3He recycle	$\approx 18.1\%$	BR-OUT-2
burner self-bred tritium	$\approx 32 \text{ kg fpy}^{-1}$ (31 % burned in situ)	MV-CX-29
burner net ^3He demand (plant scale)	$170.2 \text{ kg fpy}^{-1}$	BR-OUT-4

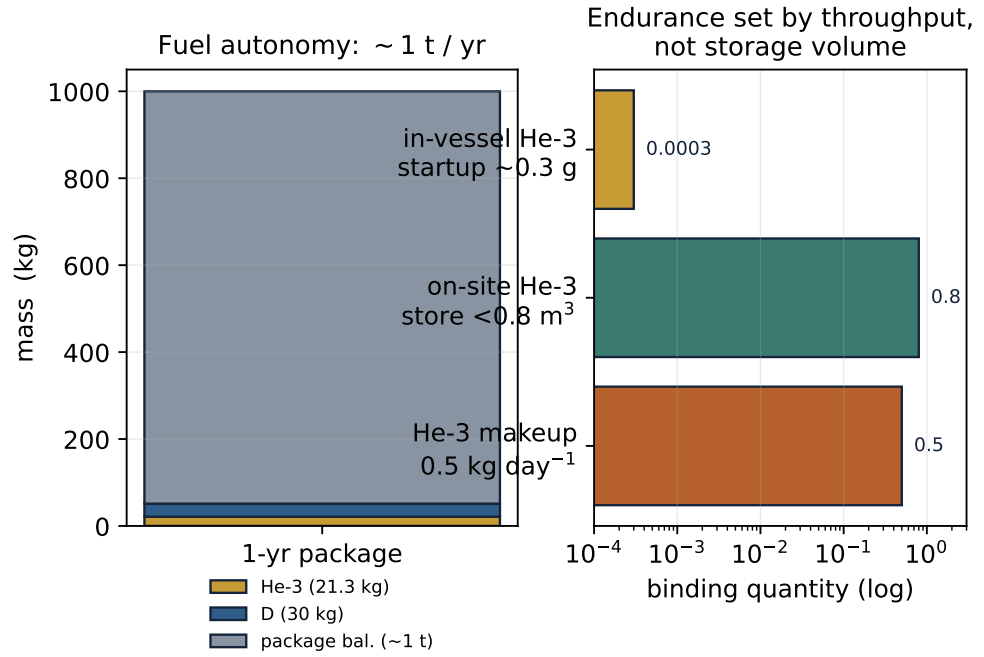


Figure 4. One-year fuel autonomy (AG-L1-A8, AG-CX-17, AG-L1-A6). Left: the ~ 1 t package is dominated by inert containment; the reactive charge is 21.3 kg ^3He + 30 kg D. Right: the binding quantities are makeup throughput and storage volume (log axis), not bulk mass.

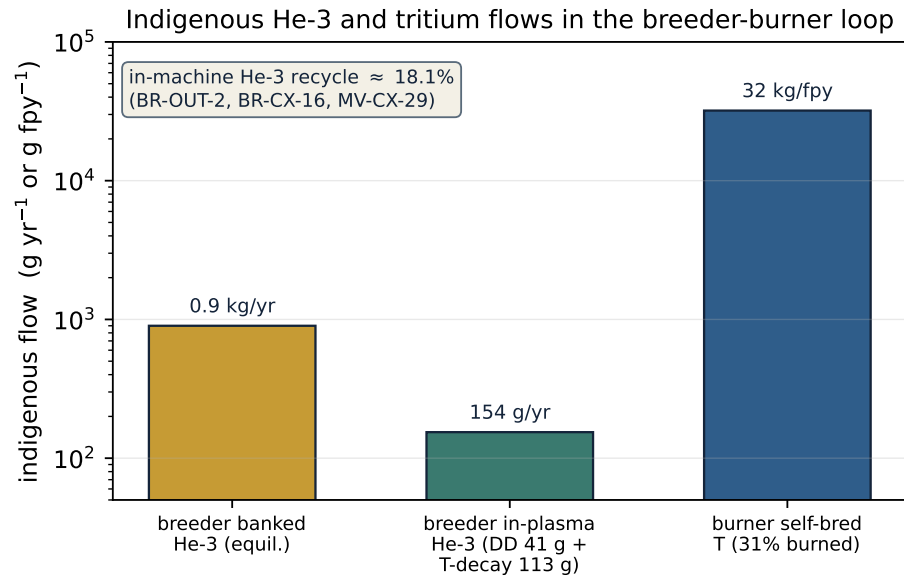


Figure 5. Indigenous helium-3 and tritium flows in the breeder–burner loop (log axis; BR-OUT-2, BR-CX-16, MV-CX-29). The breeder banks ≈ 0.9 kg yr⁻¹ of ^3He and co-produces ≈ 154 g yr⁻¹; the burner self-breeds ≈ 32 kg fpy⁻¹ of tritium and recycles $\approx 18.1\%$ of its ^3He in-machine.

4.6 Burn-lean siting knob

Figure 6 shows the burn-lean trade of equation (16): pumping bred tritium before it burns strips 60 % of the neutron power — from 234 MW down to the 93.5 MW D–D 2.45 MeV floor — for a $\approx 4\%$ loss of fusion power. A fixed installation can thus dial its neutron signature to fit a shielding and permitting envelope without a hardware change, which is exactly the freedom a base or an urban-adjacent site needs.

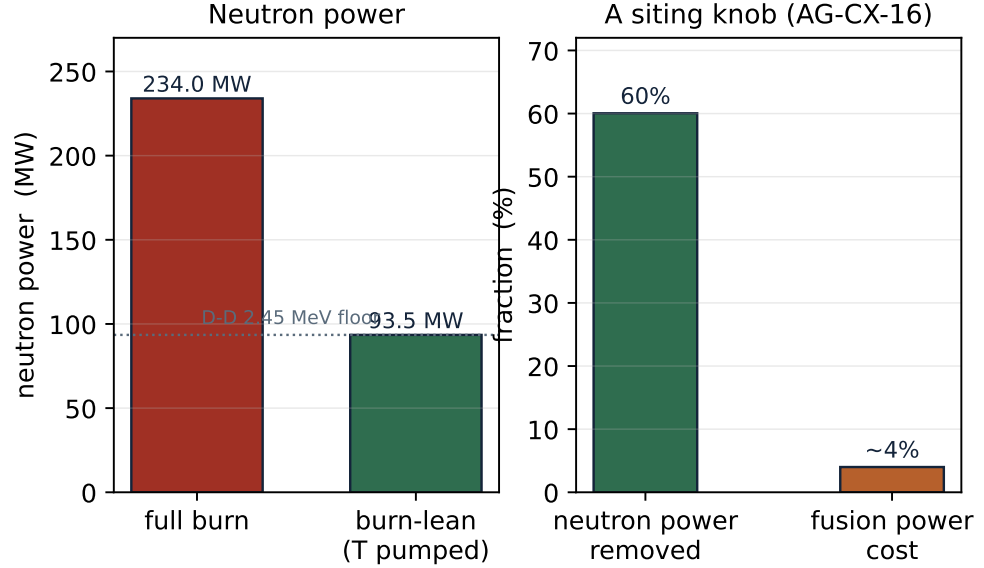


Figure 6. Burn-lean neutron suppression (AG-CX-16). Left: neutron power falls from 234 MW to the 93.5 MW D–D floor when bred tritium is pumped before it burns. Right: the trade removes 60 % of the neutron power for a $\approx 4\%$ fusion-power cost.

4.7 Fleet closure and the lunar gate

Figure 7 evaluates the indigenous closure of equation (15). A plant-scale burner's $170.2 \text{ kg fpy}^{-1}$ net helium-3 demand requires $\sim 45\text{--}190$ breeders per burner across the plausible breeding-ratio range $\text{TBR} = 1.186\text{--}1.80$. The indigenous flows therefore seed and partly sustain the loop but do not close it at deployable ratios; fleet-scale helium-3 is lunar-gated on a late-2030s schedule (BR-OUT-4), consistent with the strategic-supply canon. Near-term units run on bankable, tritium-derived and self-bred helium-3.

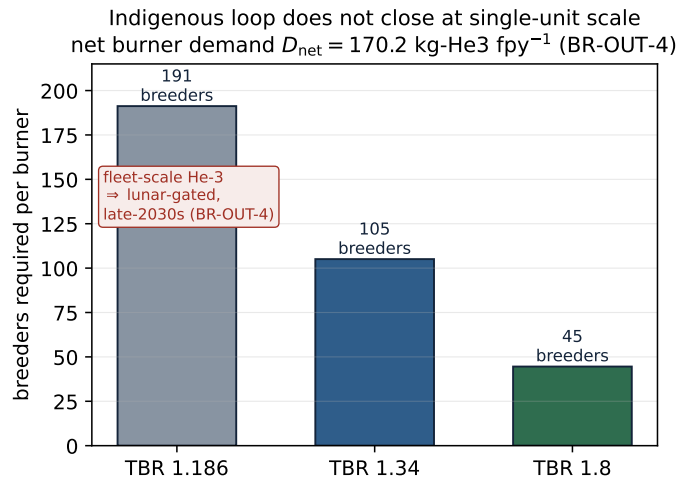


Figure 7. Fleet closure (BR-OUT-4). Breeders required per burner to meet the $170.2 \text{ kg fpy}^{-1}$ net demand, as a function of the breeder's net tritium-breeding ratio. The indigenous loop does not close at single-unit scale; fleet-scale ^3He is lunar-gated.

4.8 The mirror physics behind the operating point

Figure 8 plots the two confinement relations of §2.1: the loss-cone half-angle $\theta_{lc} = \arcsin R_m^{-1/2}$ narrows as the 26.49 T plug raises the mirror ratio, and the ambipolar potential $e\phi_c/T_e = \ln(n_{\text{plug}}/n_{\text{cell}})$ reaches 2.77 at the design density ratio of 16. These are the physics gates that the frozen power balance closes and on which the islanded envelope rests.

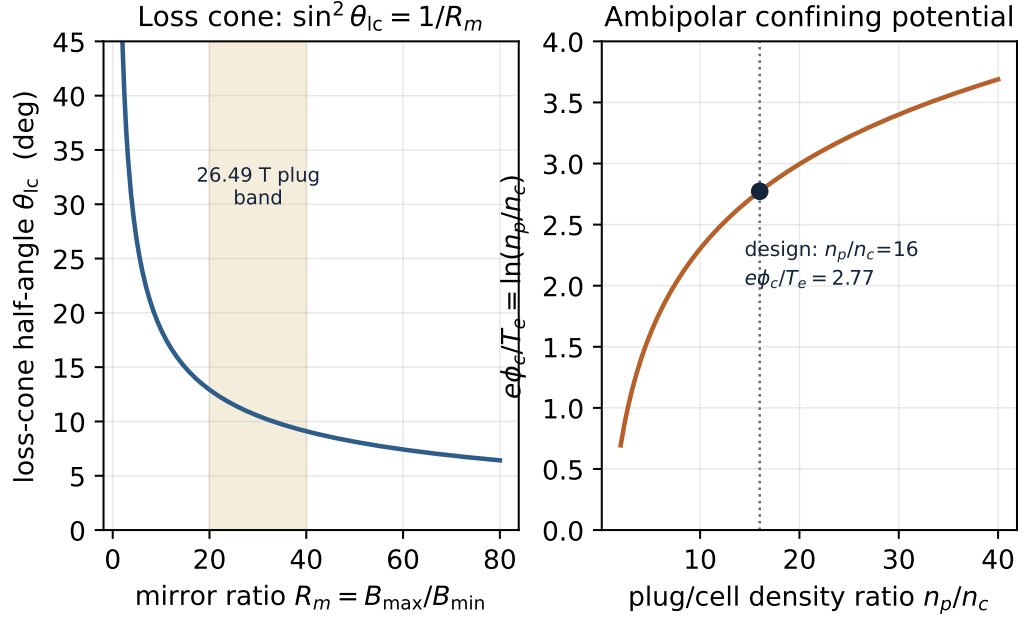
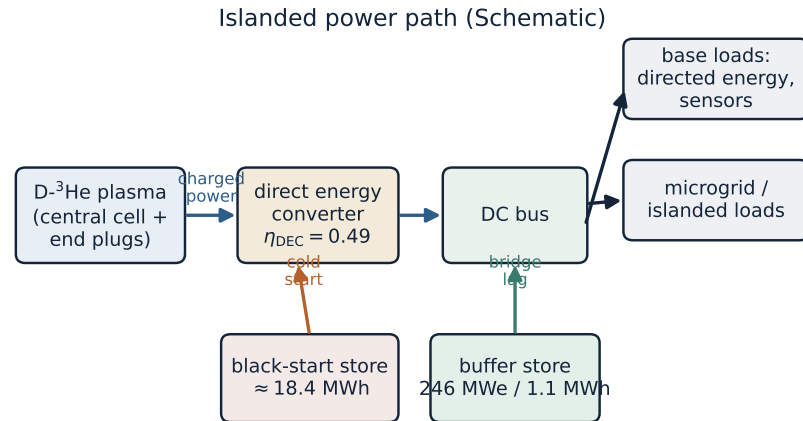


Figure 8. Tandem-mirror confinement relations at the operating point. Left: loss-cone half-angle versus mirror ratio, $\sin^2 \theta_{lc} = 1/R_m$; the 26.49 T plug band is marked. Right: ambipolar Boltzmann potential $e\phi_c/T_e = \ln(n_{\text{plug}}/n_{\text{cell}})$, with the design ratio $n_{\text{plug}}/n_{\text{cell}} = 16$ giving $e\phi_c/T_e = 2.77$.

4.9 The islanded power path

Figure 9 assembles the elements into the islanded power path: charged power flows from the plasma through the DEC to a DC bus, which serves the base loads directly while the buffer store bridges the plasma lag and the black-start store cold-starts the plant. There is no rotating thermal machine in the primary electrical path, which is what permits the millisecond trim and the deep, grid-forming load-following demonstrated above.



ms-class DEC trim + buffer bridges the ~ 8 s plasma lag; black-start store cold-starts with no grid (AG-CX-20, AG-SX-01)

Figure 9. Islanded power path of a fixed Aegis installation (*Schematic.*) Charged power is direct-converted ($\eta_{\text{DEC}} = 0.49$) to a DC bus serving base loads; a 246 MWe / 1.1 MWh buffer bridges the plasma lag and an ≈ 18.4 MWh store provides black-start (AG-CX-20, AG-SX-01).

5 Discussion

Against the tandem-mirror literature. The operating point sits within the physics envelope established by the classical tandem-mirror program [1, 2] and revived by the simplified high-gain configuration of Fowler, Moir and Simonen [5, 6]. The ambipolar-potential gate of equation (2), with $e\varphi_c/T_e \approx 2.77$ at $n_{\text{plug}}/n_{\text{cell}} = 16$, is squarely in the Pastukhov–Cohen regime [3, 4]; the required plug over-density is specified by the frozen design, not assumed. The HTS axisymmetric-mirror results of WHAM [8] and the sustained hot-electron confinement of the gas-dynamic trap [9] are the contemporary experimental context that makes a high-field plugged mirror credible. Our contribution is not a new confinement claim but the coupling of this operating point to a quantified islanded electrical and fuel-cycle envelope.

Against the direct-conversion literature. The $\eta_{\text{DEC}} = 0.49$ composite efficiency on the 14.7 MeV proton channel is consistent with the venetian-blind and multi-grid electrostatic converters analysed since Moir and Barr [7]; the collector and architecture trades that support it are treated in companion papers. The operationally decisive point for islanded service is not the peak efficiency but the *absence of a thermal time constant*: because equation (6) is realised electrostatically, the trim of equation (8) is available at power-electronics speed, which is what enables the deep load-following of figure 2.

Against the microgrid literature. The islanded case is posed exactly as a grid-forming microgrid problem [12, 13, 14]: a dispatchable source, an energy buffer sized to the source’s slowest actuator, and a black-start store. The buffer of equation (9), at 1.1 MWh and 246 MWe, is small precisely because the DEC is fast and only the ~ 8 s plasma lag must be bridged; this is a far lighter storage requirement than a thermal plant of comparable rating would impose [15]. Posing the throttle bound as a control-barrier constraint [16] inside a reference-governor law [17] places the load-following inside a certified safe set, in the same spirit as learned tokamak control [18] but with an explicit forward-invariance guarantee.

Against the helium-3 literature. The strategic reading of the fuel cycle follows Wittenberg–Santarius–Kulcinski [10, 11]: terrestrial helium-3 is scarce ($1\text{--}2\text{ kg yr}^{-1}$ domestic flow), the breeder’s banked tritium-derived helium-3 ($\approx 0.9\text{ kg yr}^{-1}$ per unit) is a strategic co-product rather than a closure mechanism, and fleet-scale demand (170.2 kg fpy^{-1} per burner) is met from lunar supply on a late-2030s schedule. The fleet-closure arithmetic of equation (15) makes the gap explicit: $\sim 45\text{--}190$ breeders per burner is not a deployable indigenous ratio, so the lunar gate is a genuine programmatic dependency, not a rhetorical one.

Companion results. This paper sits in the Kronos 2026 series. The burner physics core and its Q_E -closure gate are established in the Aegis forward-base burner paper [doi:10.5281/zenodo.22645693; <https://www.kronosfusionenergy.com/publications/aegis-burner.html>] and the recirculation-resolved closure [doi:10.5281/zenodo.22645860; <https://www.kronosfusionenergy.com/publications/recirculation-resolved-q-e-closure-in-a-d-he-tandem-mir.html>]; the direct-conversion basis in [doi:10.5281/zenodo.22645700; <https://www.kronosfusionenergy.com/publications/direct-energy-conversion.html>]; the neutron-lean siting case in [doi:10.5281/zenodo.22645917; <https://www.kronosfusionenergy.com/publications/the-neutron-lean-siting-case-for-a-d-he-tandem-mirror-b.html>]; the millisecond HVDC-native load interface in [doi:10.5281/zenodo.22645783; <https://www.kronosfusionenergy.com/publications/etrovolt-hvdc-native-power-interface.html>]; the non-closing helium-3 fuel cycle and lunar gate in [doi:10.5281/zenodo.22645734; <https://www.kronosfusionenergy.com/publications/helium-3-fuel-cycle.html>]; and the fleet tritium-breeding lever in [doi:10.5281/zenodo.22645706; <https://www.kronosfusionenergy.com/publications/blanket-tritium-breeding-lever.html>]. The present paper adds the islanded operating envelope and the fuel-cycle symbiosis that tie those results into a grid-independent installation.

6 Limitations

The one honest gate is fleet-scale helium-3 supply. Equations (14)–(15) show that the indigenous breeder–burner flows seed and partly sustain the loop but do not close a plant-scale burner’s 170.2 kg fpy^{-1} net demand at any deployable breeder-to-burner ratio; closure requires the lunar

helium-3 supply on a late-2030s schedule. Near-term units run on bankable tritium-derived and self-bred helium-3, and the burn-lean mode and the light autonomy package stand independently of the lunar schedule, but the utility-scale fleet case rests on that gated input. Separately, the islanded load-following envelope of figure 2 is a modelled result built from the frozen envelope parameters (AG-CX-20, concept-class); the DEC-trim, buffer-rating and black-start numbers are adjudicated gates, while the full closed-loop demonstration on the plant twin is the next verification step.

7 Conclusion

Aegis operates as a grid-independent fixed installation, and the case is quantitative rather than aspirational. At the frozen operating point ($Q_E = 1.318$, $f_n = 5.44\%$, $\eta_{\text{DEC}} = 0.49$) the net electric output is near-linear in central-cell length — $+104.2/+850.0/+2831.9$ MWe at 55/440/1400 m — at a first-wall load of 1.17 MW m^{-2} . Grid-independence closes end to end: a 0–104 MWe millisecond-class islanded load-following depth inside a certified stability bound, a 246 MWe / 1.1 MWh buffer that bridges the 8 s plasma lag, an ≈ 18.4 MWh black-start store, and a ~ 1 t one-year fuel package whose binding quantity is a 0.5 kg day^{-1} makeup throughput. The fuel closes a breeder–burner helium-3 symbiosis — breeder-banked tritium decaying to helium-3 ($\approx 0.9 \text{ kg yr}^{-1}$ per unit), $\approx 18\%$ in-machine recycle, and $\approx 32 \text{ kg fpy}^{-1}$ of self-bred tritium — seeded indigenously and scaled from lunar supply, and a burn-lean mode removes 60 % of the neutron power for $\approx 4\%$ of fusion power, giving a tunable neutron signature. These properties follow directly from the aneutronic, direct-conversion, linear-mirror architecture, and together they constitute a defensible, auditable basis for grid-independent fusion power at a defense installation.

Acknowledgments

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Reproducibility. The islanded-operation, black-start, fuel-autonomy and helium-3-flow results are frozen gates in the Kronos Physics De-Risking Register (KX-L1-A9, AG-CX-20, AG-SX-01, AG-L1-A8, AG-CX-17, AG-L1-A6, BR-OUT-2, BR-CX-16, MV-CX-29, AG-CX-14, AG-CX-16, BR-OUT-4) and are regenerable from the frozen 0-D mirror power-balance solver, the Kronos toolkit fuel-cycle model, the Monte-Carlo neutronics and the microgrid dispatch simulator, together with the figure-generation script deposited with this paper. This paper is archived at doi:10.5281/zenodo.22645862 and its official version is at <https://www.kronosfusionenergy.com/publications/islanded-forward-base-operation-of-the-aegis-d-he-tande.html>; the register is at doi:10.5281/zenodo.22133057.

Data availability

This paper is archived at doi:10.5281/zenodo.22645862. The islanded-envelope, black-start, fuel-autonomy and helium-3-flow inputs used for figures 1–9 are archived with the deposit and reference the Kronos 2026 series; all quantitative values trace to the register (doi:10.5281/zenodo.22133057) by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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